

METHODOLOGY DOCUMENTATION

Leave-One-Galaxy-Out Cross-Validation

Toy-object optimisation using 11 visually selected "cleaner" S4G galaxies

Calibration of SEP and MTOjects foreground-masking parameters for subsequent application to the full 182-galaxy S4G sample

Study	S4G foreground-object removal
Calibration sample	11 visually reviewed, lower-contamination galaxies
Validation design	Leave-one-galaxy-out cross-validation (LOGO-CV)
Injection design	Two independent deterministic six-toy realisations per galaxy
Prepared	26 August 2026

Executive summary

This methodology uses galaxy-level leave-one-out cross-validation to calibrate foreground-masking parameters on a deliberately restricted sample of 11 S4G galaxies judged by visual review to contain relatively few pre-existing foreground objects. The images are “cleaner”, not contaminant-free. Synthetic foreground-like toy objects provide known truth for measuring recovery while monitoring collateral masking of the galaxy image.

Recommended methodological description

Leave-one-galaxy-out cross-validation with an independent synthetic-injection validation set, followed by full-sample refitting and deployment to the complete 182-galaxy S4G sample.

For each of 11 folds, one galaxy is excluded and the optimiser searches for parameters using the remaining 10. The fold-derived candidate is evaluated on the excluded galaxy and is also evaluated across all 11 galaxies using a second, independently seeded toy realisation. After candidate selection, each algorithm is re-optimised using all 11 calibration galaxies. The resulting frozen parameter set is then applied uniformly to the full set of 182 science images.

1. Purpose and scientific rationale

The purpose is to select defensible global parameters for SEP and MTOjects without treating visibly contaminated science images as if they were clean ground truth. Real S4G images contain genuine galaxy structure, instrumental artefacts and foreground or background sources. Synthetic injection supplies an exact object-level truth mask while retaining the authentic spatially varying galaxy background.

- Calibration target: recovery of injected foreground-like objects.
- Protection target: limit excessive or incorrect masking of the underlying science image.
- Independent unit: the galaxy, not the individual toy pixel or toy object.
- Deployment target: one frozen parameter set applied consistently to all 182 S4G galaxies.

2. Calibration sample

The calibration sample contains the following 11 galaxies, selected after visual review for substantially lower foreground contamination than the wider sample. Selection is purposive rather than random and should be reported as such.

#	Galaxy	#	Galaxy
1	IC1954	7	NGC3486
2	NGC0289	8	NGC3681
3	NGC0986	9	NGC4133
4	NGC1097	10	NGC4450
5	NGC1367	11	NGC7531
6	NGC2903		

Interpretation

The sample supports calibration for lower-contamination S4G fields. It does not, by itself, establish generalisation across every morphology, inclination, surface-brightness regime or foreground density represented among all 182 galaxies.

3. Synthetic injection design

Each original galaxy image receives two immutable and reproducible toy-object realisations. Both SEP and MTObjects read the same materialised injection payloads, truth masks and checksums, ensuring that algorithm comparisons are paired rather than confounded by different toy placements.

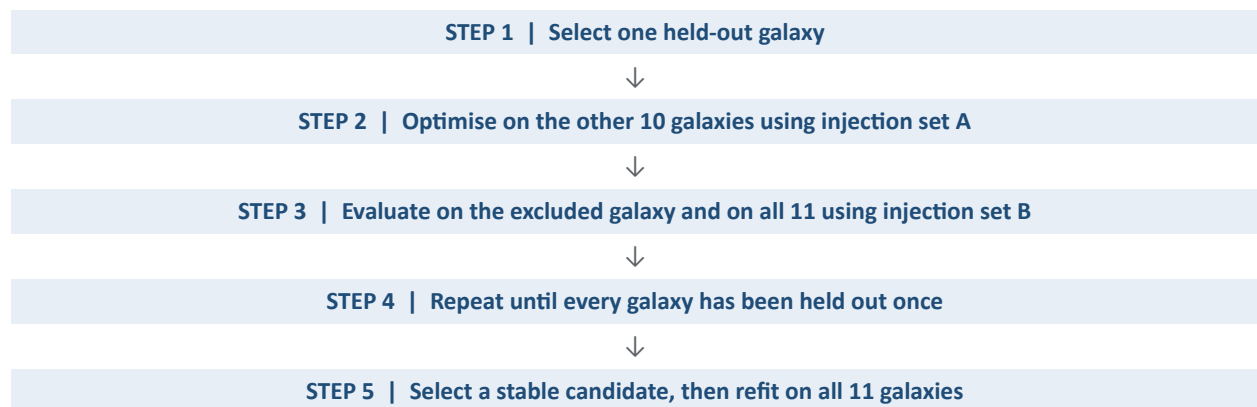
Injection set	Purpose	Per galaxy	Total toys
Cross-validation	Optimisation within each 10-galaxy training fold	6 toys	66
Winner selection	Independent realisation for comparing fold-derived candidates	6 different toys	66
Combined	Injection-realisation coverage across the experiment	12 toys	132

The two sets use different global random seeds. Consequently, toy positions, object classes, widths, axis ratios, position angles and peak amplitudes can differ. Independence is therefore achieved with respect to toy realisation, not with respect to the underlying galaxy sample.

- Six toys are injected into each galaxy in each set.
- Toy peaks span 6–30 times the robust image-noise estimate.
- Truth masks are dilated by one pixel for recovery scoring.
- The cross-validation and winner-selection seeds are 202608261 and 202608262, respectively.
- Payload checksums protect the paired comparison from accidental regeneration or mismatch.

4. LOGO-CV fold construction

Leave-one-galaxy-out cross-validation is the leave-one-group-out form of cross-validation, with each complete galaxy treated as one group. Eleven folds are constructed after a deterministic shuffle. Each galaxy appears once as the held-out case and appears in the training set for the other ten folds.



With $N = 11$ galaxies, each fold contains 10 training galaxies and one held-out galaxy. Six held-out toys are scored per fold, producing 66 toy-level held-out observations across the full LOGO-CV cycle. These toy observations are clustered within galaxies; they must not be reported as 66 independent galaxies.

5. Optimisation within each fold

Each fold runs a separate Optuna parameter search. Eight startup trials provide broad exploration and 32 adaptive Tree-structured Parzen Estimator trials refine the search, giving 40 trials per fold and 440 fold-training trials per algorithm. Four image workers are used for computation.

1. Prepare the 10 training galaxies with their fixed cross-validation injection payloads.
2. Run the candidate foreground-masking algorithm on every injected training image.
3. Measure recovery of the known truth masks and the incremental mask introduced beyond detections already present in the original image.
4. Combine recovery and masking terms into the algorithm-specific objective, including penalties or feasibility constraints.
5. Retain the best training candidate from the 40 trials for subsequent held-out assessment.

Why the galaxy is the fold unit

Pixel-level or toy-level random splitting would leak the same galaxy background into training and validation. Holding out the complete galaxy tests transfer to a different image background and morphology.

6. Validation and candidate selection

6.1 Held-out assessment

The best candidate from each training fold is applied to the excluded galaxy. Combining the 11 held-out outcomes estimates how the optimisation procedure transfers across the cleaner-galaxy sample. Because each fold validates on one galaxy, individual fold scores may be volatile; the aggregate distribution and cross-fold stability are more informative than a single fold.

6.2 Independent injection-realisation selection

Each of the 11 fold-derived candidates is also evaluated across all 11 galaxies using the winner-selection injection set. This prevents candidate selection from depending solely on the six particular toys used during fold training. The aggregate all-11 objective is based primarily on mean recovery and masking measures, not the median of the 11 held-out results.

6.3 Algorithm-specific rule

Algorithm	Candidate-selection rule
SEP	Choose the fold-derived candidate with the best aggregate all-11 objective on the independent injection realisation.
MTOjects	First require the candidate to satisfy recovery and masking feasibility conditions; then choose the feasible candidate with the best aggregate all-11 objective.

Not nested cross-validation

This is best described as LOGO-CV plus independent injection-realisation selection. A fully nested design would reserve every outer held-out galaxy exclusively for final assessment while performing all parameter and candidate selection inside an inner loop.

7. Full-sample refitting and deployment

Cross-validation estimates stability and transfer; it does not deliberately discard one galaxy from the final calibration. Once the assessment stage is complete, SEP and MTOjects are each re-optimised using all 11 cleaner galaxies. These final all-11 parameters become the production configurations.

Deployment then freezes the selected parameters and applies them without per-galaxy retuning to all 182 S4G images. The deployment stage should preserve provenance: algorithm version, parameter JSON, detection-image mode, injection-manifest checksum, clean-list version, random seeds and run timestamp.

Stage	Galaxies	Purpose	Permitted adaptation
Fold training	10 per fold	Parameter search	Yes, within the fold
Held-out assessment	1 per fold	Estimate transfer	No
Winner selection	All 11; alternate toys	Compare fold candidates	Selection only
Final refit	All 11	Production parameters	Yes, final optimisation
Science deployment	All 182	Foreground masking	No per-galaxy retuning

8. Quantities to report

- Galaxy-level sample size: 11 calibration galaxies.
- Fold structure: 11 folds, each with 10 training galaxies and one held-out galaxy.
- Toy counts: 66 toys in each injection set; 132 toys across both realisations.
- Held-out recovery: report the mean, median, range and preferably a galaxy-level bootstrap confidence interval across the 11 held-out galaxy outcomes.
- Parameter stability: report the distribution or range of major parameter values across fold winners.
- Mask burden: report mean and maximum incremental masked fraction, including any feasibility threshold breaches.
- Algorithm comparison: use the paired galaxy/injection design and compare component metrics rather than assuming SEP and MTOjects objective values are directly interchangeable.

9. Assumptions and limitations

- The 11 galaxies were selected visually and are not a random sample of the 182; selection bias is therefore possible.
- "Cleaner" means lower apparent contamination, not verified absence of real foreground objects.

- Additional toy seeds improve sampling of injection placements but do not increase the number of independent galaxies.
- The second injection set is independent of the first toy realisation but uses the same galaxy backgrounds.
- LOGO-CV with only 11 galaxies can have high variance. Aggregate performance and fold stability should be emphasised.
- Application to all 182 galaxies is deployment, not proof that the parameters are equally optimal for every morphology or contamination regime.
- A later external validation set containing independently labelled real foreground objects would provide stronger evidence of scientific generalisability.

10. Suggested thesis wording

Methods paragraph

Foreground-masking parameters were calibrated using leave-one-galaxy-out cross-validation on 11 visually selected S4G galaxies with comparatively low foreground contamination. In each of 11 folds, parameters were optimised on 10 galaxies containing fixed synthetic foreground-like injections and assessed on the excluded galaxy. Fold-derived candidates were additionally compared across all 11 galaxies using a second deterministic injection realisation generated with a different random seed. Following cross-validation, each algorithm was refitted on the complete 11-galaxy calibration sample, and the resulting frozen parameters were applied uniformly to the full 182-galaxy S4G dataset. Galaxies, rather than individual toys or pixels, were treated as the independent sampling units.

Results caveat

The procedure evaluates transfer within a purposively selected lower-contamination galaxy sample. The independent injection realisation reduces sensitivity to particular toy placements but does not constitute an independent galaxy validation sample.

11. Reproducibility record

Item	Implemented value
Clean-galaxy list	IC1954; NGC0289; NGC0986; NGC1097; NGC1367; NGC2903; NGC3486; NGC3681; NGC4133; NGC4450; NGC7531
Fold seed	202608260
Cross-validation injection seed	202608261
Winner-selection injection seed	202608262
Toys per galaxy per set	6
Toy peak range	6-30 robust sigma
Truth dilation	1 pixel
Trials per fold	40: 8 startup + 32 adaptive TPE
Workers	4

Item	Implemented value
Detection image	Original science image
Algorithms	SEP and MTOjects
Deployment sample	182 S4G galaxies

References and further reading

Hastie, T., Tibshirani, R. and Friedman, J. (2009). The Elements of Statistical Learning: Data Mining, Inference, and Prediction, 2nd ed. Springer. Chapter 7 discusses model assessment and cross-validation.

scikit-learn developers. Cross-validation: evaluating estimator performance.

https://scikit-learn.org/stable/modules/cross_validation.html

scikit-learn developers. LeaveOneOut.

https://scikit-learn.org/stable/modules/generated/sklearn.model_selection.LeaveOneOut.html